Causal sufficient dimension reduction for studying environmental mixtures

An applied introduction to CSDR and the `csdr` R package

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Why causal dimension reduction?

What do we want to learn from an environmental mixture?

Scientific goals

- Estimate the **joint health effect**
- Learn the **exposure-response curve**
- Identify **important contributors**
- Find a useful **mixture summary**

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Highly correlated exposures, sparse support, and confounding—all at once.

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Different methods prioritize different scientific goals.

Mixtures methods make different tradeoffs

Method	Dimension	Description	Natural strength
WQS / qgcomp	$d = 1$	Scalar joint quantile shift	Interpretable effect
PCA / PCP	$1 \leq d \leq p$	Unsupervised dimension reduction	Pattern discovery
BKMR / BART	$d = p$	Full exposure-response	Nonlinear interactions
CSDR	**learn** $1 \leq d \leq p$	**Supervised causal dimension reduction**	**Flexible causal summary**

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Key distinction

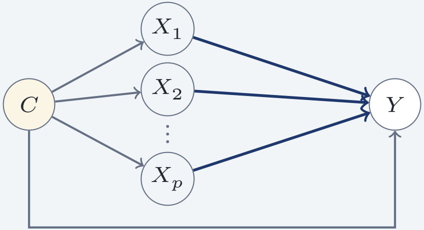
CSDR estimates a **supervised causal dimension reduction** of the exposure mixture X that preserves the **exposure-response curve**.

The joint health effect is a causal inference problem

Outcome regression (common approach)

$$\mathbb{E}(Y \mid X = x, C = c) = h(x) + \gamma^\top c$$

Flexible $h(x)$, linear-additive $\gamma^\top c$



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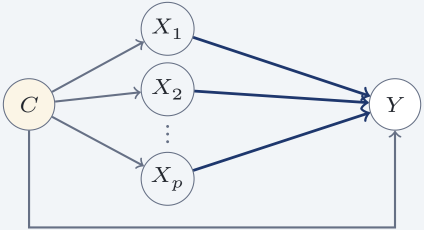

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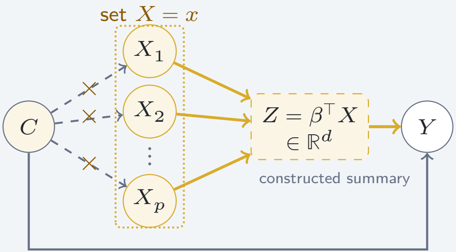

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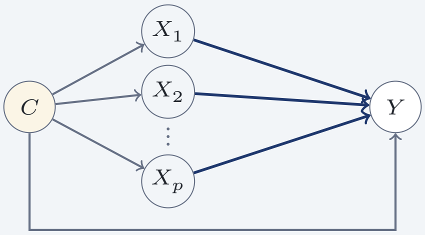

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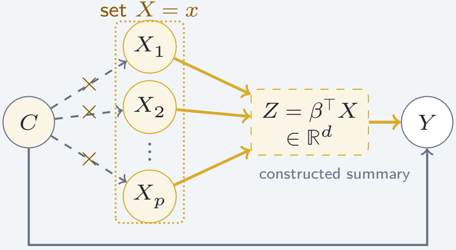


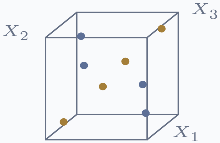
average over C

The exposure-response curve $\mu(x)$ is what we truly care about—not $\mathbb{E}(Y \mid X, C)$.

Dimension reduction makes mixtures estimable and interpretable

Full mixture

$d = p$



The eight points
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Single score

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A single score collapses
one response pattern.

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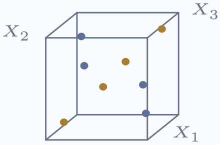

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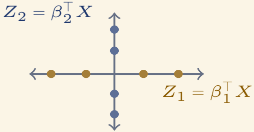

Two directions retain both patterns with denser support.

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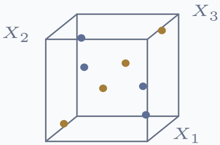

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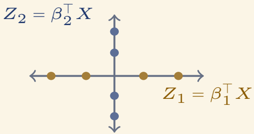

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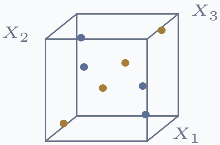

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WHY IT HELPS

Multiple patterns
more flexible than a single mixture score

Denser support
more nearby observations; greater statistical power

Easier estimation
lower-dimensional nuisance and response models

Clearer results
easier to visualize and communicate

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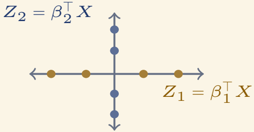

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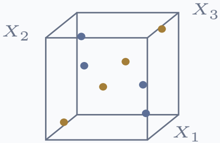

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Causal inference and dimension reduction are both important for mixtures...how does CSDR put them together?

CSDR is causally supervised dimension reduction

Main Assumption

$$\mu(x) = \mathbb{E}(Y^x) = g(\beta^\top x)$$

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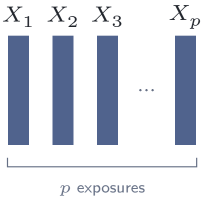
$X \in \mathbb{R}^p$



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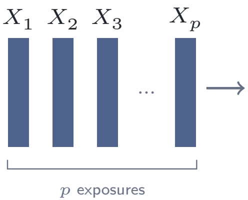
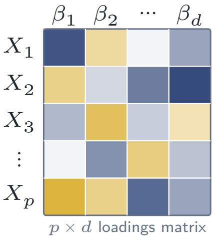


2. Causal directions

$\beta \in \mathbb{R}^{p \times d}$



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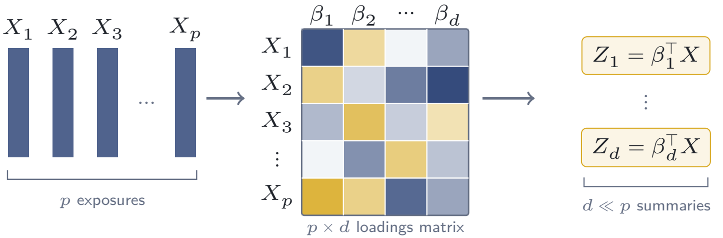

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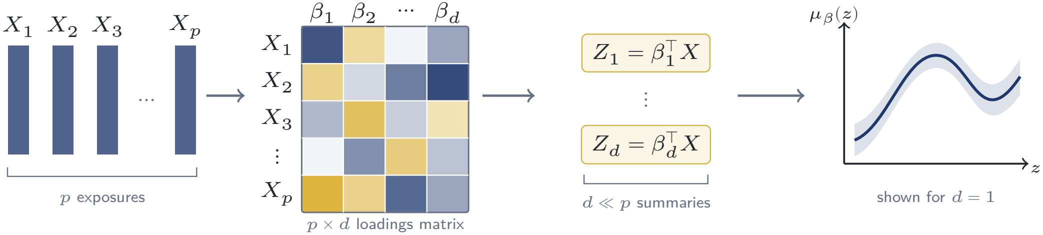

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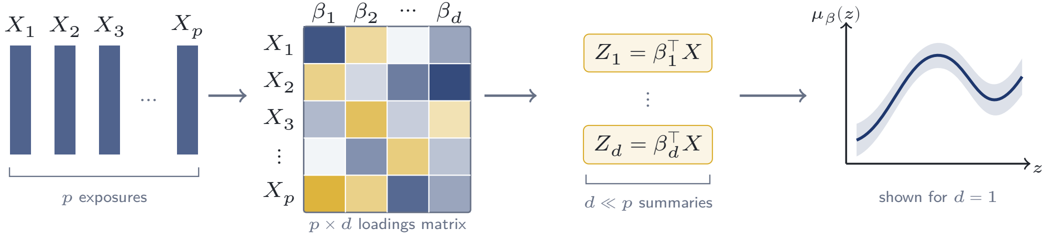
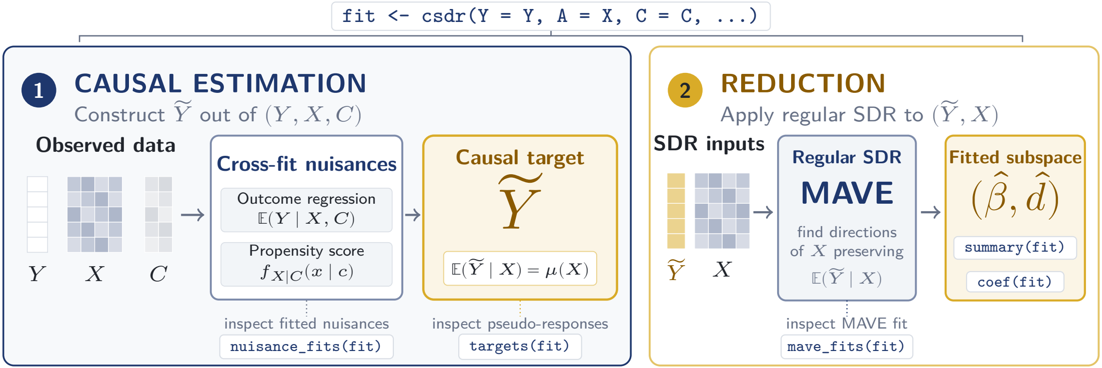


◎ Plain-language target

Replace p exposures with d summaries that **preserve the exposure-response curve** $\mu(x)$.

How CSDR estimates the reduced exposure



How CSDR estimates the reduced exposure

```
fit <- csdr(Y = Y, A = X, C = C, ...)
```

1 CAUSAL ESTIMATION

Construct $\widetilde{Y}$ out of (Y, X, C)

Observed data

Y X C →

Cross-fit nuisances

Outcome regression
$\mathbb{E}(Y \mid X, C)$

Propensity score
$f_{X|C}(x \mid c)$

→

Causal target

$\widetilde{Y}$

$\mathbb{E}(\widetilde{Y} \mid X) = \mu(X)$

inspect fitted nuisances `nuisance_fits(fit)`

inspect pseudo-responses `targets(fit)`

2 REDUCTION

Apply regular SDR to $(\widetilde{Y}, X)$

SDR inputs

$\widetilde{Y}$ X →

Regular SDR

MAVE

find directions of X preserving $\mathbb{E}(\widetilde{Y} \mid X)$

→

Fitted subspace

$(\widehat{\beta}, \widehat{d})$

`summary(fit)`

`coef(fit)`

inspect MAVE fit `mave_fits(fit)`

AFTER FITTING

downstream analysis

reduced exposure

$\widehat{Z} = \widehat{\beta}^{\top} X$

`Z = scores(fit)`

⟶

exposure-response curve

$\widehat{\mu}_{\widehat{\beta}}(z)$

`estimate_ers(Y = Y, A = Z, C = C, ...)`

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Why not interpret β like PCA loadings?

Not unique: βR spans the same subspace.

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Basis $\beta^{(A)}$ **Basis $\beta^{(B)}$**

$$\begin{bmatrix} 1.00 & 0.00 \\ 0.00 & 0.71 \\ 0.00 & -0.71 \end{bmatrix} \xleftrightarrow{p=3,\ d=2} \begin{bmatrix} 0.80 & -0.60 \\ 0.42 & 0.57 \\ -0.42 & -0.57 \end{bmatrix}$$

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$$\lambda_j = \left[\beta(\beta^\top \beta)^{-1}\beta^\top\right]_{jj}$$

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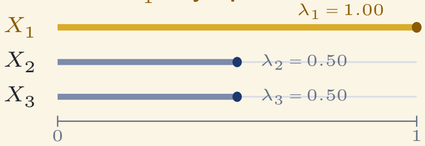

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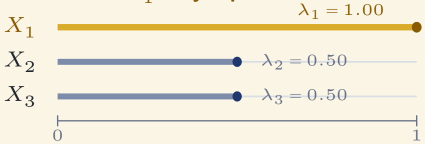

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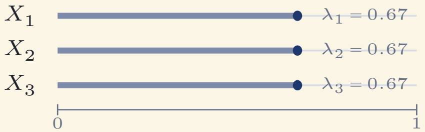

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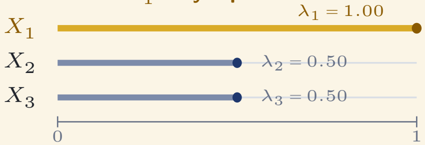

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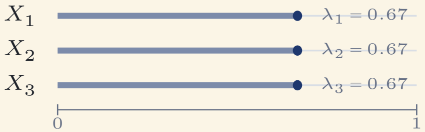


SIS PROPERTIES

- $0 \le \lambda_j \le 1$ — higher = more represented
- $\sum_{j=1}^{p} \lambda_j = d$ — additive decomposition
- **Basis-invariant** — stable to full-rank transforms of β

What does the researcher learn?

Back to our scientific goals...

1) Estimate the joint health effect

$$\widehat{\Delta} = \hat{\mu}(x^{\text{high}}) - \hat{\mu}(x^{\text{ref}})$$

Compare causal means under two mixture profiles.

2) Learn the exposure-response curve

$$\hat{\mu}_{\beta}(z) = \hat{g}(z)$$

Fit the exposure-response curve across reduced Z.

3) Identify important contributors

$$\hat{\lambda}_1, \dots, \hat{\lambda}_p$$

SIS measures how strongly each exposure is represented in the causal subspace.

4) Find a useful mixture summary

$$\widehat{Z} = \hat{\beta}^{\top} X \in \mathbb{R}^{\hat{d}}$$

Replace p exposures with learned scores that preserve the exposure-response curve.

Practical FAQ about CSDR

1) Which outcomes?	**Binary, count, or continuous Y** Use outcome-appropriate nuisance learners.
2) What is linear?	**Nuisances and $g(\beta^\top x)$ are nonparametric** Only $Z = \beta^\top X$ is linear. CSDR accommodates BART, XGBoost, neural network, etc.
3) How are interactions studied?	**Nonlinear g can encode interactions among X** For $d \geq 2$, visualize $g(z_1, z_2)$; interpret interactions using supported X_j, X_k contrasts.
4) Are CIs available?	**Analytic pointwise 95% CIs are available for $\widehat{\mu}_{\widehat{\beta}}(z)$** Inference for $\widehat{\beta}$, $\widehat{\lambda}$, and $\hat{d}$ is available only through bootstrapping.
5) How long does it take?	**Runtime depends on n, p, and the nuisance learners** The more computationally intensive the nuisance fit, the longer the full CSDR takes.

Simulations and Real Data

CSDR improves the estimated exposure-response curve

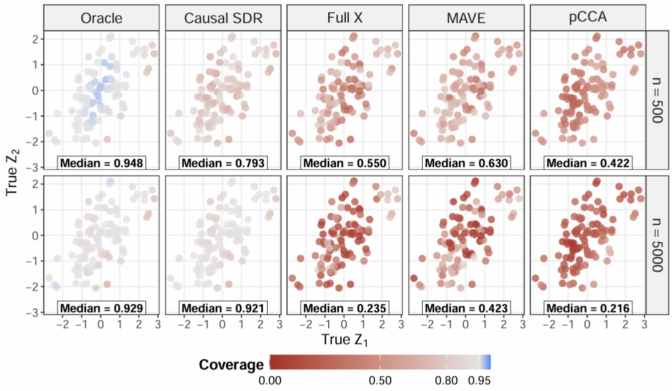
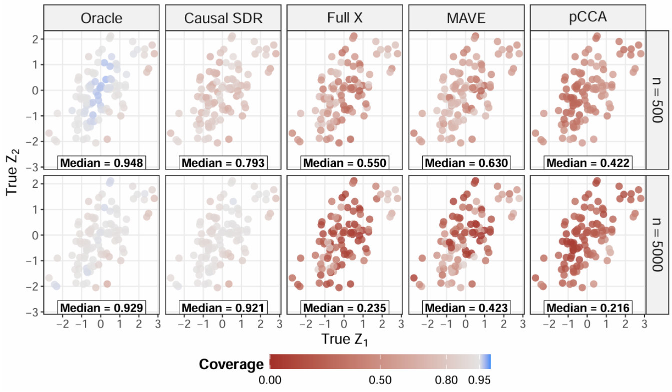


Figure 1: Pointwise 95% coverage for the exposure-response curve ($p = 10$, $d = 2$); methods differ only in β estimation.

CSDR improves the estimated exposure-response curve



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Simulation message: CSDR approaches the oracle and outperforms Full X (no dimension reduction), MAVE (no confounding adjustment), and pCCA (only linear confounding adjustment). Both causal adjustment and dimension reduction matter.

PFAS application: four exposures summarized in one direction

We use the **Atlanta African American Maternal-Child Cohort** $(n = 305)$ to study the effect of a PFAS mixture $(p = 4)$ on infant birth-weight. Covariates include maternal age, BMI, education, tobacco use, marijuana use, and infant sex.

ESTIMATED REDUCTION $(\widehat{d} = 1)$

$$\hat{Z} = 0.77\,\mathrm{PFOS} + 0.14\,\mathrm{PFOA} + 0.26\,\mathrm{PFNA} - 0.56\,\mathrm{PFHxS}$$

SUBSPACE IMPORTANCE SCORE $\widehat{\lambda}_j$

PFOS	0.60
PFOA	0.02
PFNA	0.07
PFHxS	0.31

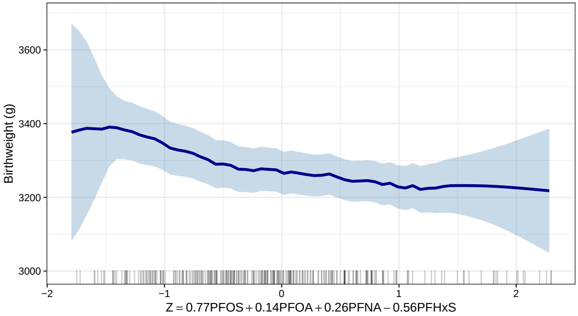
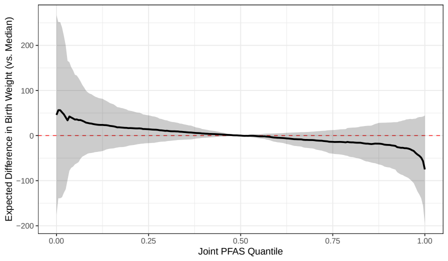


Figure 2: PFAS exposure-response curve from ATL-AA

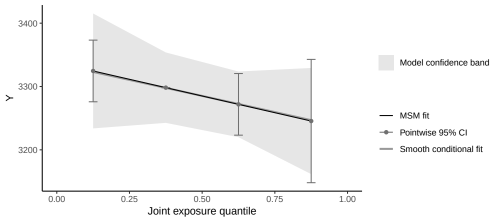
BKMR and qgcomp comparison analysis

BKMR



All four PFAS share a quantile versus the median.

qgcomp



All four PFAS shift together across exposure quartiles.

Takeaway: Different estimands, but in general, increasing PFAS leads to decreasing birthweight.

The `csdr` workflow

Setup outcome, exposures, and confounders

R code

```r
library(csdr)
data(csdr_example2)

Y <- csdr_example2$Y
A <- csdr_example2$A
C <- csdr_example2$C

head(Y, 5)
head(A[, c(1:3, 18:20)], 5)
head(C, 5)
```

Console

```
> head(Y, 5)
     obs1      obs2      obs3      obs4      obs5
   0.3805    0.1838   -2.0486    0.4977    5.4798

> head(A[, c(1:3, 18:20)], 5)
          A1     A2     A3    A18    A19    A20
obs1 -1.066  1.573  0.953  0.033 -0.480  0.500
obs2 -1.028  0.325 -0.364 -0.967 -1.543 -0.572
obs3 -0.800 -0.103 -0.433 -0.011  0.505  0.623
obs4  2.492  0.582 -1.134 -1.473 -1.138  1.551
obs5  1.733 -0.114  0.615  1.083  0.350  0.133

> head(C, 5)
          C1    C2     C3     C4     C5
obs1 -0.935 1.312 -0.294 -1.979  1.150
obs2 -0.694 0.531  0.367 -2.259 -1.293
obs3 -0.548 0.712 -0.088 -0.407  1.828
obs4  0.766 1.050 -0.498 -0.593 -0.604
obs5  1.852 -0.554  0.849  0.643 -0.312
```

Fit the causal subspace

R code

```
sl_lib <- c(
  "SL.glm", "SL.earth", "SL.ranger"
)

learners <- csdr_learners(
  sl_library = sl_lib
)

set.seed(42)
fit <- csdr(
  Y = Y, A = A, C = C,
  L = 5, seed = 42,
  learners = learners,
  max_dim = 5,
  keep_nuisance = TRUE
)

summary(fit)
```

Console

```
> summary(fit)
Summary of causal sufficient dimension reduction fit

 variant d_hat   n p mave_method target_exposure
      DR     2 100 20    meanMAVE               A

Learners:
 outcome: SuperLearner regression
   family: gaussian
   SL.library: SL.glm, SL.earth, SL.ranger
 gps: MVN GPS with linear conditional mean
   method_gps: linear; density_floor: 1e-16

Target construction:
Effective folds: 5
PO marginalization: crossfit
```

Inspect the fitted subspace and scores

R code

```r
B_hat <- coef(fit)
Z_hat <- scores(fit)

mave <- mave_fits(fit)
dim_selection <- mave$dimension_selection

B_hat[c(1, 2, 20), , drop = FALSE]
sis(fit)
head(Z_hat, 2)
dim_selection
```

Console

```
> B_hat[c(1, 2, 20), , drop = FALSE]
       dir1    dir2
A1  -0.3401  0.5777
A2  -0.3524  0.4195
A20 -0.0521 0.0784

> sis(fit)
    A1    A2    A3    A4    A5    A6    A7    A8    A9   A10
 0.464 0.312 0.533 0.390 0.000 0.004 0.002 0.020 0.001 0.021
   A11   A12   A13   A14   A15   A16   A17   A18   A19   A20
 0.068 0.059 0.028 0.020 0.006 0.005 0.015 0.036 0.006 0.009

> head(Z_hat, 2)
       score1  score2
obs1  -0.1862  0.4515
obs2   1.5224 -0.3484

> dim_selection
Dimensions      0    1    2    3    4    5
CV-value     1.58 1.13 0.75 0.77 0.92 0.97
Selected dimension: 2
```

Estimate the response in the reduced exposure

R code

```r
z_grid <- matrix(
  rep(apply(Z_hat, 2, median), each = 50),
  nrow = 50,
  dimnames = list(NULL, colnames(Z_hat))
)
z_grid[, 1] <- seq(
  quantile(Z_hat[, 1], 0.05),
  quantile(Z_hat[, 1], 0.95),
  length.out = 50
)

ers <- estimate_ers(
  Y = Y, A = Z_hat, C = C,
  a_eval = z_grid, estimator = "DR",
  L = 5, seed = 42,
  verbose = FALSE
)

head(ers$results)
```

Console

```
> head(ers$results)
  score1 score2 estimate     se ci_lower ci_upper effective_n
1 -1.990 0.0015    2.676 0.5599    1.579    3.774       4.502
2 -1.912 0.0015    2.693 0.4762    1.760    3.627       4.559
3 -1.834 0.0015    2.703 0.3920    1.935    3.472       4.671
4 -1.756 0.0015    2.703 0.3136    2.088    3.317       4.851
5 -1.678 0.0015    2.687 0.2493    2.198    3.176       5.113
6 -1.601 0.0015    2.652 0.2100    2.241    3.064       5.476
    min_gps h_used_1 h_used_2
1 4.049e-05    0.578    0.596
2 4.049e-05    0.578    0.596
3 4.049e-05    0.578    0.596
4 4.049e-05    0.578    0.596
5 4.049e-05    0.578    0.596
6 4.049e-05    0.578    0.596
```

Conclusion

When should I use CSDR?

Use CSDR when you want

- A joint causal exposure-response curve
- Supervised dimension reduction with estimated d
- Flexible modeling of complex relationships
- Variable importance measures

CSDR does not replace

- Good study design
- Confounder selection
- Adequate exposure support
- Sensitivity analysis

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One-sentence takeaway

CSDR learns the smallest exposure representation $Z = \beta^\top X$ that preserves the exposure-response curve, and `csdr` turns that target into an applied workflow.

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Questions?

CSDR preprint

`arxiv.org/abs/2606.14840`

`csdr` R package

`github.com/tXiao95/csdr`

Appendix

Identification assumptions

Assumptions A1–A5 from the CSDR framework

A1. Identification for X

a) Consistency: $X = x \Longrightarrow Y = Y^x, \quad \forall x,$
b) Strong positivity: $\inf_{x,c} \pi(x \mid c) \geq \varepsilon > 0,$
c) Conditional ignorability: $Y^x \perp\!\!\!\perp X \mid C, \quad \forall x.$

A2. Smoothness for continuous exposures

$m(\cdot, c), \pi(\cdot \mid c) \in C^3(\mathcal{X})$, with uniformly bounded derivatives; the conditional variance and its x-derivatives are uniformly bounded. The kernel k is bounded, differentiable, symmetric, and second order:

$$\int k(u)\,du = 1, \qquad \int u\,k(u)\,du = 0, \qquad \int u^2 k(u)\,du < \infty.$$

A3. Existence of the CCMS

Let

$$\mathcal{A} = \left\{ S : \begin{array}{c} \mu(x) = \mu(P_S x) \\ \text{for all } x \in \mathcal{X} \end{array} \right\}.$$

Then

$$\mathcal{S}_{E(Y^x)} = \bigcap_{S \in \mathcal{A}} S \in \mathcal{A}.$$

A4. Well-defined reduced exposure

$$\beta^\top x = \beta^\top x' \quad \Longrightarrow \quad Y^x = Y^{x'} \text{ a.s.}$$

A5. Strong positivity for $Z = \beta^\top X$

$$\inf_{z,c} \pi_\beta(z \mid c) \geq \varepsilon > 0.$$

Identification consequence. Under A1–A5, Z inherits consistency and strong positivity, as well as conditional ignorability, and $\mu(x) = \mathbb{E}\{m(x, C)\}$, $\mu_\beta(z) = \mathbb{E}\{m_\beta(z, C)\}$.

Additional estimation assumptions

Assumptions A6–A8 from the CSDR framework

A6. Additive confounding model

There exist measurable functions g, h, f such that

$$Y = g(\beta^\top X) + h(C) + \varepsilon_Y,$$
$$\mathbb{E}(\varepsilon_Y \mid X, C) = 0,$$
$$X = f(C) + \varepsilon_X,$$
$$\mathbb{E}(\varepsilon_X \mid C) = 0, \quad \varepsilon_X \perp\!\!\!\perp C.$$

A8. Uniform nonsingularity

For

$$\mathcal{X}_{h,i}(\beta, x) = \begin{pmatrix} 1 \\ h^{-1}\beta^\top(X_i - x) \end{pmatrix},$$

define the local design matrix

$$S_n(\beta, x) = \frac{1}{n}\sum_{i=1}^{n} K_{h,i}(x)\,\mathcal{X}_{h,i}(\beta, x)\mathcal{X}_{h,i}(\beta, x)^\top.$$

Assume

$$\sup_{\beta, x}\|S_n(\beta, x)^{-1}\| = O_p(1).$$

A7. csMAVE regularity conditions

(a) X has compact support; its marginal density f_X has bounded fourth derivative and is bounded away from zero.

(b) $\mu(x) = g(\beta^\top x)$, where g has bounded, continuous third derivatives.

(c) K is a spherically symmetric density with bounded derivative and bounded support. All moments exist, and

$$\int u u^\top K(u)\, du = I.$$

(d) With $V(z) = \partial g(z)/\partial z$,

$$\mathbb{E}[V(\beta_0^\top X)V(\beta_0^\top X)^\top] \quad \text{is nonsingular.}$$

(e) The first-stage ERS estimator satisfies

$$\max_{1\le i\le n} |\widehat{\mu}(X_i) - \mu(X_i)| = O_p(R_n), \qquad R_n = o(h^2).$$

Theorem 1: recovering the causal subspace

Theorem 1 (Convergence of csMAVE)

Let β_0 be a basis for the true CCMS $\mathcal{S}_{\mathbb{E}(Y^x)}$, and let $\widehat{\beta}$ minimize the csMAVE objective using an estimated ERS response $\widehat{\mu}(X)$. Define

$$\delta_n = \left\{\frac{\log n}{nh^p}\right\}^{1/2}, \qquad \max_{1\leq i\leq n} |\widehat{\mu}(X_i) - \mu(X_i)| = O_p(R_n).$$

Under Assumptions A1–A3, A7, and A8, if

$$h \to 0, \qquad d \geq d_0, \qquad \frac{nh^p}{\log n} \to \infty,$$

then

$$\boxed{\left\|(I_p - \widehat{\beta}\widehat{\beta}^\top)\beta_0\right\| = O_p(h^3 + h\delta_n + h^{-1}R_n)\,.}$$

Theorem 2: learning the structural dimension

Theorem 2 (Consistency of $\hat{d}$)

For each candidate dimension $d \in \{0, 1, \dots, p\}$, let $\hat{\beta}_d$ be the corresponding csMAVE estimate and let $\mathrm{CV}(d)$ be the cross-validated prediction error for $\hat{\mu}(X_i)$ using the reduced exposure $\hat{\beta}_d^\top X_i$. Select

$$\hat{d} = \operatorname*{arg\,min}_{d \in \{0,1,\dots,p\}} \mathrm{CV}(d).$$

Under Assumptions A1–A3 and A7,

$$\boxed{\Pr(\hat{d} = d_0) \longrightarrow 1 \qquad \text{as } n \to \infty.}$$

Choosing among causal response constructions

Construction	Main nuisance inputs	Practical role
RA	Outcome regression	Lower-cost sensitivity analysis
DR	Outcome regression + GPS	Package default; robust to one correctly specified nuisance component
PO	Outcome regression + GPS	Doubly robust but noisier SDR response
RP	Models for $E(Y \mid C)$ and $E(A_j \mid C)$	Efficient under additive-confounding structure